

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: pSU2 neo, PAG 60
Species Used	Human, pancreatic cell lines		

Before the Pulse

Cell Growth Medium	Daigo's T (10% Fetal Calf Serum)	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	Not given

Wash Solution Hanks' Balanced Salt Solution

The Pulse

Instruments Used Gene Pulser ® apparatus

Electroporation Temperature Room temperature
Electroporation Medium* Hanks' Balanced Salt Solution

Cuvette Gap Not given

Cell Density 1.5 x 10⁶ (6) / ml

Voltage 0.50 to 0.7 kV

Volume of Cells 1 ml

Field Strength Not given

DNA Concentration 1 to 10 µg / ml

DNA Resuspension Buffer Hanks' Balanced Salt Solution

Capacitor 25 µF

Volume of DNA 1 ml

Resistor (Pulse Controller) Ω none

After the Pulse

Time Constant 0.9 msec

Outgrowth Medium Daigo's T (5% FCS)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 37 °C
Length of Incubation not given
Selection Method or Assay Used pSV2 Neo (G418)
Electroporation Efficiency 1 to 20 clones /1µl DNA; depends on the cell line
Per Cent Survival 50%

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